

QuickDraw — trained bases vs classical baselines

1. Main result

We introduce a parametric quantum-circuit family of unitary image bases with $O(N \log N)$ forward / inverse cost and $O(\log N)$ trainable parameters per dimension. **(i)** On line-drawing imagery (QuickDraw, 32×32), the trained block-wrapped basis exceeds every linear classical baseline tested — including the dataset-fitted optimum BlockPCA-8 — by **+5.97 dB** PSNR over BlockDCT-8 at $\rho = 0.20$, after ≈ 5 minutes of training on a single RTX 3090. **(ii)** On natural images (DIV2K, 256×256), the same

family matches BlockDCT-8 to within ≈ 0.3 dB, saturating the **Ahmed–Natarajan–Rao limit** under which the KLT of a stationary Gaussian AR(1) source converges to the DCT. **(iii)** Accordingly, the parametric basis offers a rigorous improvement over BlockDCT on image distributions whose patch statistics depart from stationary Gaussian AR(1), and parity on those that do not.

2. Setup + pipeline

$$m = n = 5, \quad d = 2^m \cdot 2^n = 32 \times 32 = 1024$$

$$N_{\text{train}} = 500, \quad N_{\text{test}} = 50, \quad \text{seed} = 42, \quad \rho \in \{0.05, 0.10, 0.15, 0.20\}$$

Per image $x \in [0, 1]^d$ with basis Φ :

$$y = \Phi x \quad (\text{forward})$$

$$k = \lfloor \rho \cdot d \rfloor, \quad y'_i = \begin{cases} y_i & \text{if } |y_i| \in \text{top-}k \\ 0 & \text{else} \end{cases} \quad (\text{compression})$$

$$\hat{x} = \text{clip}(\Phi^{-1}y', 0, 1), \quad \text{MSE} = \frac{1}{d} \|x - \hat{x}\|_2^2, \quad \text{PSNR} = 10 \log_{10}(1/\text{MSE})$$

For block transforms: same pipeline applied independently to each 8×8 tile, top- k pooled across all blocks.

AR(1) (first-order autoregressive). Each pixel = $\rho_{\text{AR}} \times \text{previous} + \text{small Gaussian noise}$:

$$x_n = \rho_{\text{AR}} x_{n-1} + \varepsilon_n, \quad \varepsilon_n \sim \mathcal{N}(0, \sigma_\varepsilon^2)$$

$\rho_{\text{AR}} \rightarrow 1$ Gaussian $\rightarrow$ KLT collapses to DCT (Ahmed–Natarajan–Rao). Empirical lag-1 autocorrelation $\hat{\rho}_{\text{AR}} = \frac{1}{2}(\hat{\rho}_{\text{row}} + \hat{\rho}_{\text{col}})$ across multiple samples: **DIV2K natural images** (4 different scenes, centre 256×256 crops) cluster in $\hat{\rho}_{\text{AR}} \in [0.84, 0.99]$ — close to the image-row-like regime, $\rho \approx 1$, where BlockDCT is near-optimal; **QuickDraw drawings** (3 samples, single 28×28 each) cluster tightly at $\hat{\rho}_{\text{AR}} \approx 0.70$ — further from the AR(1)–Gaussian limit, leaving room for a trained basis to beat BlockDCT. The two distributions barely overlap, exactly the QuickDraw vs DIV2K asymmetry the results table shows. Distinct from the keep-ratio ρ above.

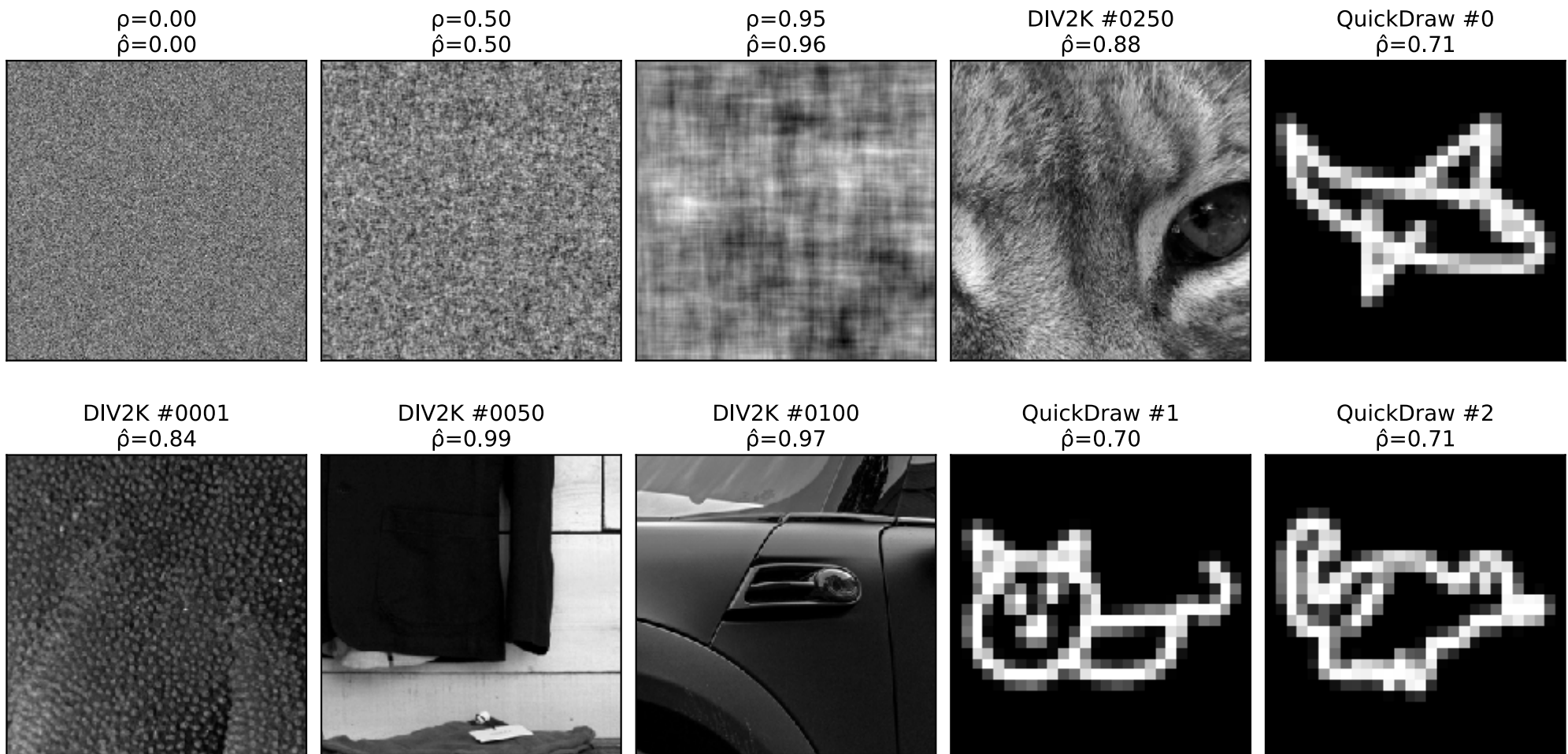


Figure 1: **Top row:** three synthetic AR(1)–Gaussian fields at $\rho = 0, 0.5, 0.95$ (sanity-checking $\hat{\rho}$); DIV2K 250 centre crop; QuickDraw drawing 0 **Bottom row:** three more DIV2K samples (1, 50, 100) and two more QuickDraw drawings (1, 2) — confirming the $\hat{\rho}_{\text{AR}}$ values are robust across samples and the two datasets cluster at different points on the AR(1) axis. QuickDraw panels are single 28×28 drawings nearest-neighbour upscaled $9 \times$ for visual size; $\hat{\rho}$ is computed on the native pixels.

3. Fits — what is Φ for each method?

method	Φ source	Φ shape	forward $y =$	compression $y' =$	inverse $x =$
bd_pca (bilateral 2D-PCA)	U, V from SVDs of column- and row-stacked centered images: $\Sigma_{\text{col}} = \frac{1}{NW-1} \sum_i (X_i - (X)) (X_i - (X))^T (H \times H)$ $\Sigma_{\text{row}} = \frac{1}{NH-1} \sum_i (X_i - (X))^T (X_i - (X)) (W \times W)$ fit on $N = 500$ images at $H = W = 32$ ($NH = NW = 16000$ samples per axis)	U is $H \times H = 32 \times 32$, full rank V is $W \times W = 32 \times 32$, full rank (each axis has 16000 $\gg$ 32 samples)	$Y = U^T(X - (X))V$, shape $H \times W$	top- k on Y : $Y_{ij} \cdot \mathbb{1}[Y_{ij} \geq \tau_k]$, $k = \lfloor \rho \cdot HW \rfloor = \lfloor \rho \cdot d \rfloor$ (same rate as DCT)	$UY'V^T + (X)$
block_bd_pca_8 (block bilateral 2D-PCA)	U_b, V_b are separable column + row eigenbases at the 8×8 patch level, fit on $N_p = 8000$ pooled patches (500 images $\times 4 \times 4 = 16$ blocks per image), via SVDs on $N_p b \approx 64000$ samples per axis	U_b, V_b are each $b \times b = 8 \times 8$, full rank, shared across all 4×4 blocks. Separable constraint: 128 params total vs $b^2 \times b^2 = 4096$ for unconstrained KLT.	per block: $Y_b = U_b^T(P - (P))V_b$ (shape 8×8 per block)	top- k pooled across all blocks, keep $k = \lfloor \rho \cdot 1024 \rfloor$ largest entries of the $4 \times 4 \times 8 \times 8$ tensor	per block: $U_b Y_b V_b^T + (P)$, then re-tile
dct (global)	closed-form (DCT-II): $\Phi[k, n] = \alpha_k \cos(\pi(n + 1/2)k/N)$ $\alpha_0 = \sqrt{\frac{1}{N}}$, $\alpha_{k \geq 1} = \sqrt{\frac{2}{N}}$ $N = 32$, no fit, no mean	$d \times d$ analytically	Φx (no mean)	top- k : $y_i \cdot \mathbb{1}[y_i \geq \tau_k]$ (rank: keep $i = 0, \dots, k - 1$)	$\Phi^T y'$
block_dct_8	closed-form (DCT-II) at $N = 8$: $\Phi[k, n] = \alpha_k \cos(\pi(n + 1/2)k/8)$ shared across all blocks	64×64 per block analytically tiled identically across the image	per block: Φp	top- k pooled across all blocks (rank: per-block keep $i = 0, \dots, k_b - 1$, $k_b = \lfloor \rho \cdot 64 \rfloor$)	per block: $\Phi^T y'$, then re-tile

Table 1: Concrete definition of Φ and the full pipeline (forward $\rightarrow$ compression $\rightarrow$ inverse) for each method. All four Φ are **orthogonal** ($\Phi\Phi^T = I$), so inverse = transpose. 2D = separable (apply 1D transform along rows then columns).

Top- k rule (default, headline results): keep the k coefficients with largest **[magnitude]**, set the rest to zero — pooled across all blocks for block transforms.

Rank rule (in parentheses; used by the `_rank` control variants): keep the **first** k coefficients in the basis’s natural ordering. For **PCA** this means the k **highest eigenvalue** directions (rows of V^T are sorted by descending λ); for **DCT** this means the k **lowest frequency** components (DC first, then increasing

3.1. Why DCT $\approx$ block PCA in the limit

$$\Phi_{\text{block_pca}_8} = \text{KLT}(\Sigma_b) \longrightarrow \Phi_{\text{block_dct}_8} = \text{KLT}\left(\lim_{\rho \rightarrow 1} \Sigma_{\text{AR}(1)}(\rho)\right)$$

DCT = what KLT becomes if the patch covariance is the analytic AR(1)–Gaussian limit. Empirically the two differ by only 0.68 dB on QuickDraw — natural patches are nearly AR(1)–Gaussian.

4. Results — PSNR (dB), seed=42

unblocked (full 32×32)	$\rho = 0.05$	$\rho = 0.10$	$\rho = 0.15$	$\rho = 0.20$
Trained PDFT bases (ours)				
★ qft	16.72	19.58	22.05	24.35
★ entangled_qft	16.72	19.58	22.05	24.35
★ tebd	16.72	19.58	22.06	24.36
Classical, top- k rule				
bd_pca	18.63	21.99	24.87	27.57
dct	16.13	18.44	20.33	22.03
fft	15.26	17.32	19.01	20.56
Classical, rank rule (control)				
dct_rank	13.84	15.24	16.49	17.63

8×8 block-wrapped	$\rho = 0.05$	$\rho = 0.10$	$\rho = 0.15$	$\rho = 0.20$
Trained PDFT bases (ours)				
★ rich	18.79	23.78	28.27	32.69
★ real_rich	18.80	23.80	28.30	32.67
★ blocked	18.12	22.40	26.20	30.05
Classical, top- k rule				
block_dct_8	17.20	20.70	23.72	26.63
block_bd_pca_8	17.00	20.50	23.41	26.05
block_fft_8	15.97	18.74	21.09	23.34
Classical, rank rule (control)				
block_dct_8_rank	13.50	14.81	15.78	16.97

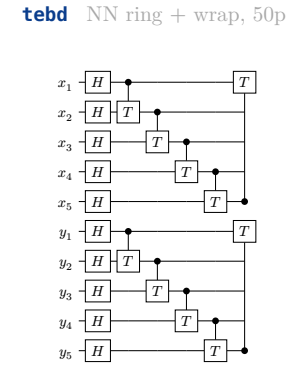
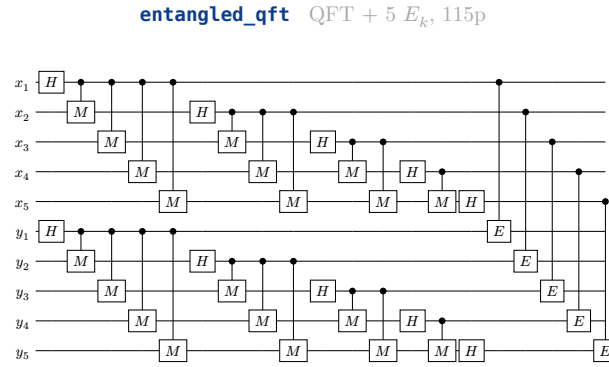
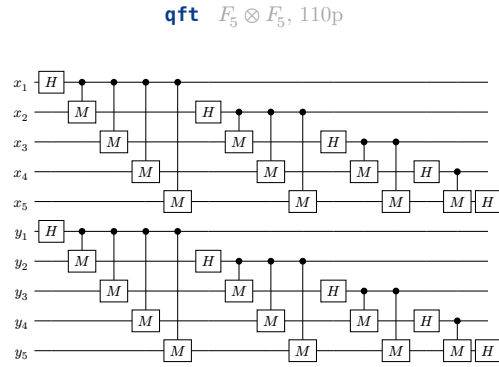
Table 2: Side-by-side: unblocked / full-image methods (left, gray header) vs 8×8 block-wrapped methods (right, light blue header). Each column groups trained PDFT bases, classical top- k baselines, and the rank-rule control. ★ = trained PDFT basis (this work); unmarked rows

5. classical baselines. Bold = best-in-group at each

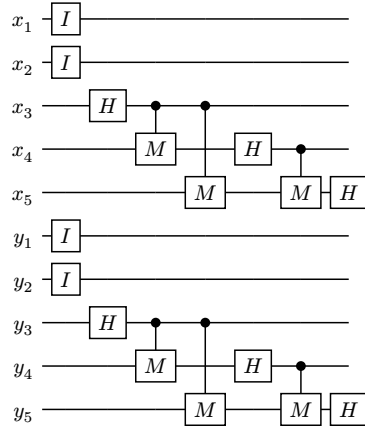
keep ratio. **real_rich** ★ wins at three of the four ratios (red highlight) using **half** the parameters of rich; rich edges it out at $\rho = 0.20$ by 0.02 dB. The two block-wrapped trained bases are essentially tied within single-seed noise; the headline emphasises **real_rich** for the parameter-count advantage. **mera** omitted (incompatible at $m + n = 10$).

5.1. Tensor networks of the trained bases at $m = n = 5$

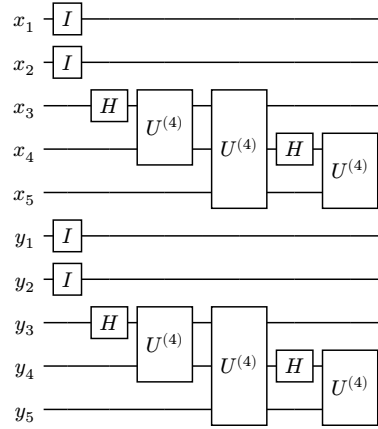
10-qubit layout (rows $x_1, \dots, x_5$ top, cols $y_1, \dots, y_5$ bottom). Seven sub-circuits arranged in a 4-column grid for side-by-side comparison. All separable except entangled_qft.



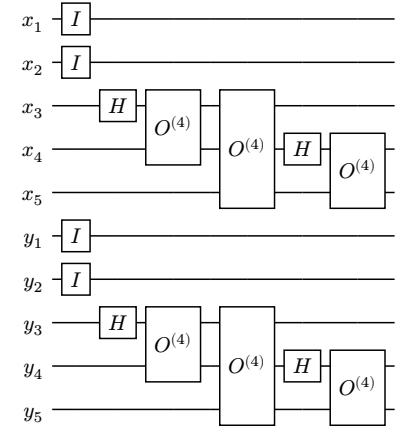
blocked $I_4 \otimes F_3$ per dim, 30p



rich $I_4 \otimes U^{\text{inner}}$, 108p



real_rich $I_4 \otimes O^{\text{inner}}$, 42p (HEADLINE)



Listing 1: Six trained-basis topologies at QuickDraw geometry ($m = n = 5$, 10 qubits per circuit). Top row — unblocked: **qft**, **entangled_qft**, **tebd** (all at the same scale and row-spacing so their 10-wire heights match exactly). Bottom row — block-wrapped: **blocked**, **rich**, **real_rich** (headline). **mera** is structurally inapplicable here ($m + n = 10 \neq 2^p$) and omitted. Param counts cover both row and col registers.

5.2. Geometry comparison at a glance

basis	topology	row-col coupling	params/dim	PSNR at $\rho = 0.20$
qft	all-to-all per dim	none	55	24.35
entangled_qft	all-to-all + 5 cross	yes (E_k)	60	24.35
tebd	NN ring per dim	none	25	24.36
blocked	outer- I + inner QFT-3	none	15	30.05
rich	outer- I + inner $U^{(4)}$	none	54	32.69
real_rich	outer- I + inner $O^{(4)}$	none	21	32.67

Table 3: All bases at QuickDraw geometry $m = n = 5$. Param counts are per dim (multiply by 2 for full transform unless row/col are tied). PSNR from Table earlier in this doc.

6. Training loss curves

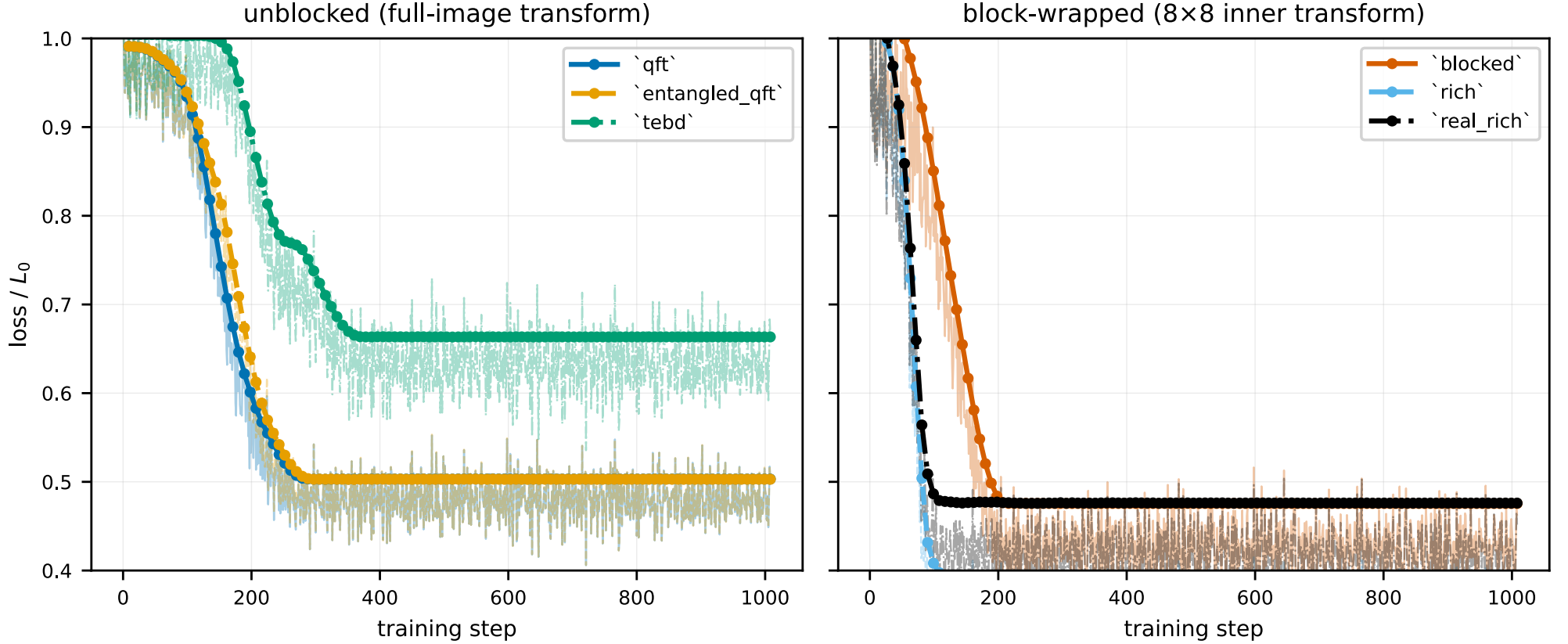


Figure 2: Per-step training loss (faint) and per-epoch validation loss (markers) for each trained basis on QuickDraw at the headline 1008-step training budget. Y-axis is **normalised** by each basis's own initial step-loss (L/L_0); cross-basis comparison is on convergence speed and floor. Left: unblocked / full-image bases. Right: 8×8 block-wrapped bases. Each basis uses a unique colour + line-style pair from a colourblind-safe palette. `mera` is omitted because $m + n = 10$ is not a power of 2 and the MERA hierarchy is structurally inapplicable.

6.1. Appendix: 2000-step plateau check

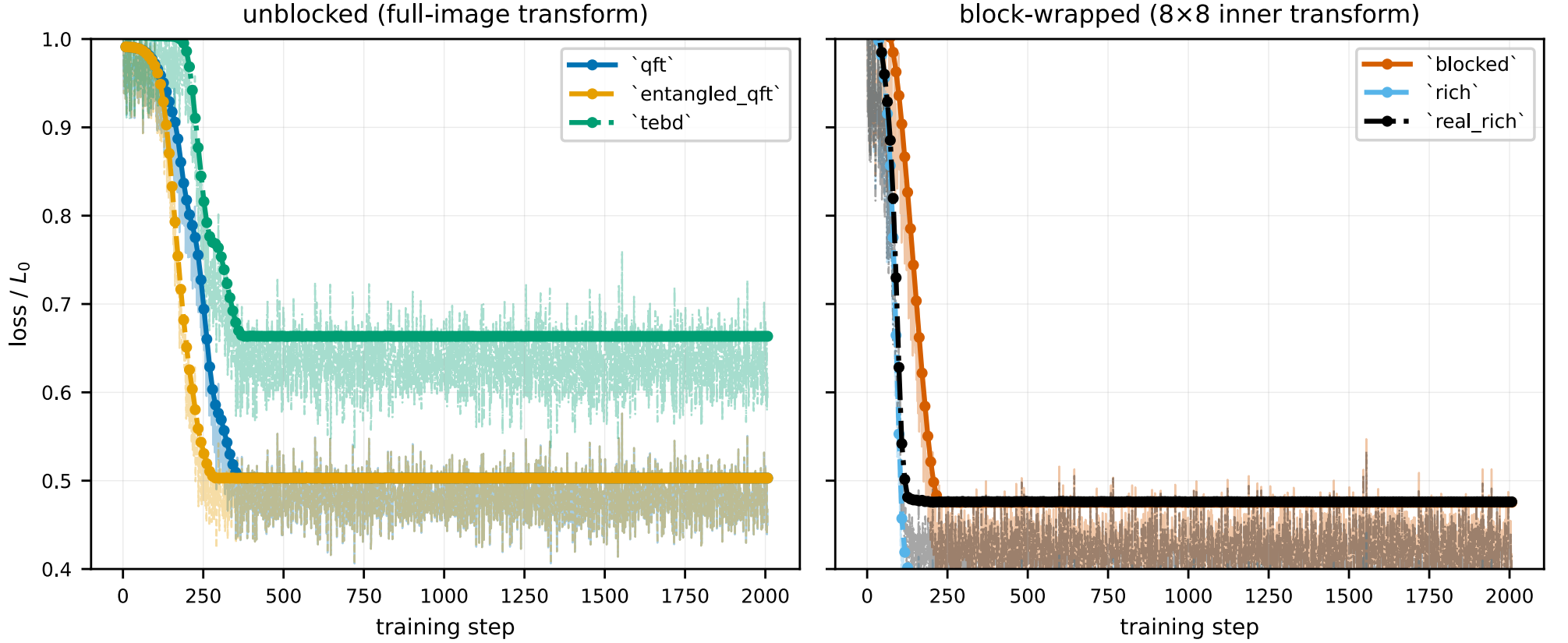


Figure 3: Same plot as the headline figure but trained for ≈ 2000 steps (epochs = 223) instead of ≈ 1000 (epochs = 112). Verifies that QuickDraw losses have plateaued by the headline budget — the windowed-mean training loss between steps 1000 and 2000 moves by ≤ 0.01 dB equivalent. Headline-table values shift by ≤ 0.05 dB between the 1000-step and 2000-step runs (*real_rich* for example: $32.67 \rightarrow 32.62$ at $\rho = 0.20$); the shifts are below the single-seed noise floor on $n_{\text{test}} = 50$.

7. Reconstructions — 2 representative images $\times$ keep ratios $\times$ bases

7.1. Image 0 — cat sketch

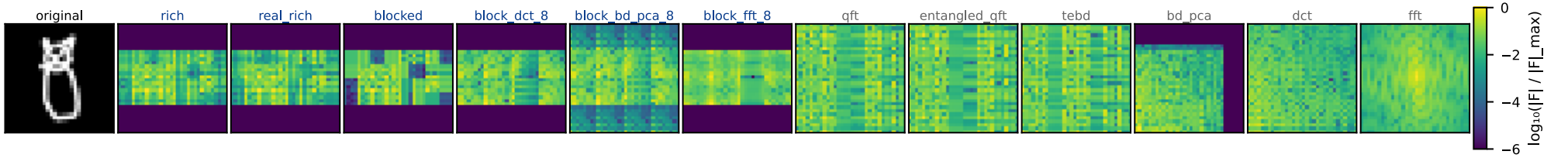


Figure 4: Frequency-space spectra (peak-normalised $\log|F|$, viridis, shared color scale) for image 0 under each basis. Same column order as the recon grid below. Note the trained block-wrapped bases (blue headers) push energy into a small number of low-coefficient cells per block — visible as the bright clusters in rich/real_rich/blocked and the band structure in block_dct_8/block_bd_pca_8. The bd_pca panel shows energy concentrated in the top-left corner (low-frequency / high-eigenvalue) along both axes — the separable-KLT analog of the DCT spectrum.

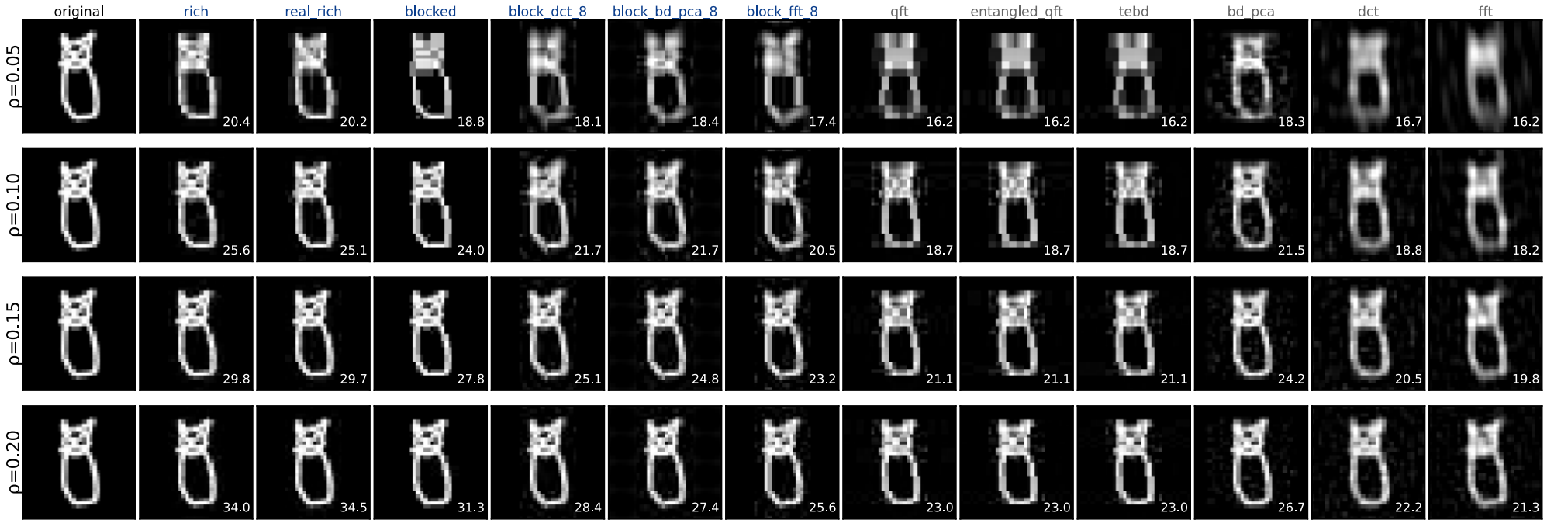


Figure 5: QuickDraw test image 0 reconstructed at four keep ratios $\rho \in \{0.05, 0.10, 0.15, 0.20\}$ (rows). Same column order as the freq-space figure above. PSNR (dB) annotated bottom-right of each cell. At $\rho = 0.20$: trained rich ≈ 34 dB vs block_dct_8 ≈ 28 dB (+6 dB) vs dct ≈ 22 dB (+12 dB).

7.2. Image 2 — fish silhouette

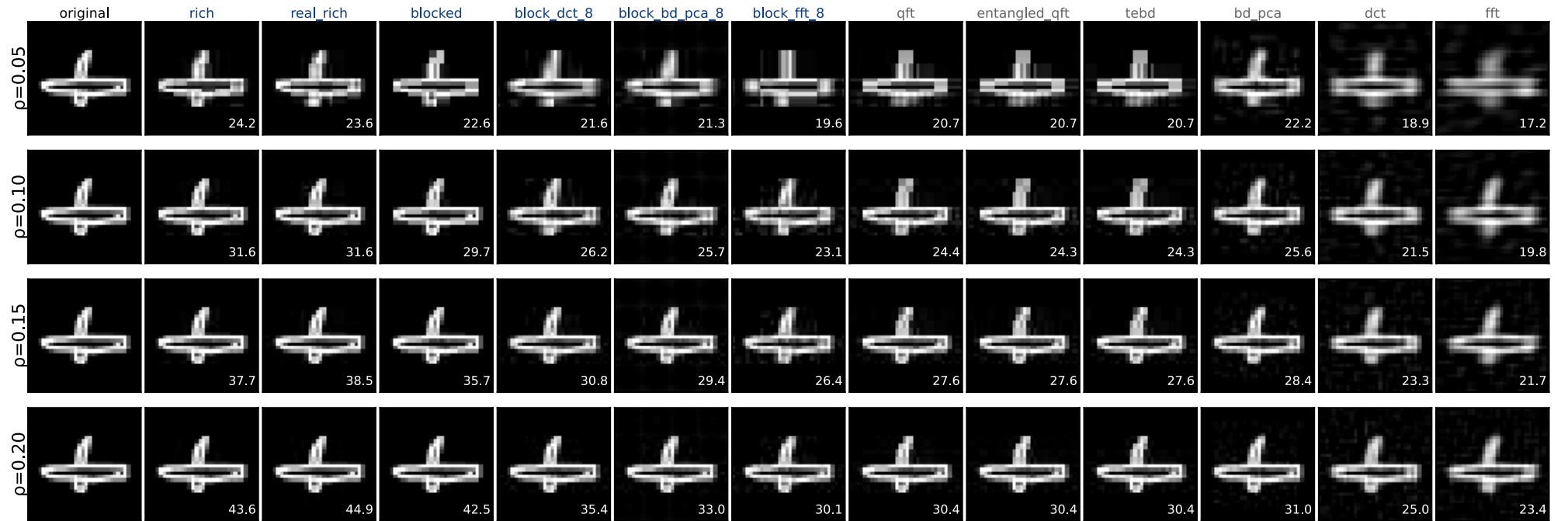


Figure 6: QuickDraw test image 2 reconstructed at the same four keep ratios. Same column layout as image 1 This image is among the strongest cases for trained tensor-network bases: at $\rho = 0.20$, trained rich ≈ 44 dB vs block_dct_8 ≈ 32 dB (+12 dB) vs dct ≈ 25 dB (+19 dB). Even at $\rho = 0.05$ (top row) rich and real_rich already preserve the recognisable fish-stroke topology while dct and fft collapse to a blurred blob.

8. Matching summary — bench name $\rightarrow$ paper figure

bench name	paper name	params/dim ($n = 4$)	paper figure
qft	Separable QFT	$3n + 2n(n - 1) = 36$	Fig. 1d, Tab. 1
entangled_qft	Entangled QFT	$36 + n = 40$	Fig. 3, Eq. (entangled_qft)
tebd	TEBD ring	$3n + 2n = 20$	App. C left, Fig. (tebd_circuit)
mera	MERA hierarchical	$3n + 4(n - 1) = 24$	App. C right, Fig. (mera_circuit)
blocked	Blocked QFT	15 ($m_{\text{in}} = 3$)	<code>\\$block_wrapper</code> , Fig. (block_basis_circuits) bottom
rich	RichBasis	54 ($m_{\text{in}} = 3$)	Fig. (block_basis_circuits) middle
real_rich	RealRichBasis	21 ($m_{\text{in}} = 3$)	Fig. (block_basis_circuits) top — headline

Table 4: Mapping from `pdft_benchmarks` basis name to the paper’s naming and circuit figure. Param counts use the conventions in `main.tex` `\$sec:qft_basis` and `\$sec:block_wrapper`.